

Smale's Mean Value Conjecture and the Pandrosion Vanishing Identity

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Abstract

We reformulate Smale's mean value conjecture as a bound on the Pandrosion quotient: $\min_{P'(c)=0} |Q(0, c)|/|Q(0, 0)| \leq (d-1)/d$, where $Q(z, w) = P(w)/w$. The classical Lagrange–Sylvester identity $\sum 1/P'(\alpha_k) = 0$ (written here in Pandrosion notation as $\zeta_P(1) = 0$), specialised to a root at the origin, yields the constraint $1 = -\sum 1/(\alpha_k R'(\alpha_k))$, from which the triangle inequality gives $\min_k |\alpha_k R'(\alpha_k)| \leq d-1$. This is a root-side bound, not a direct critical-point bound. We identify the extremal polynomial $P = z^d - z$, which attains $\min_c |P(c)/c|/|P'(0)| = (d-1)/d$ exactly for every d , and report a numerical sweep over 40,000 random polynomials ($d = 2, \dots, 9$) with no violations. Section 5 describes what an eventual proof would still require; this paper supplies structural constraints, not a full proof.

Scope statement. We do *not* claim a proof of Smale's MVC for any $d \geq 4$. Section 2 derives a *root-side* bound $\min_k |\alpha_k R'(\alpha_k)| \leq d-1$ from the classical Lagrange–Sylvester identity; Section 3 gives the critical-point identity $S(c) = |cR'(c)|$; Section 5 explains why these two ingredients alone do not close the MVC and what a completed argument would still need.

1 Smale's conjecture in Pandrosion form

Conjecture 1.1 (Smale, 1981). *For any polynomial P of degree $d \geq 2$ with $P(0) = 0$ and $P'(0) \neq 0$, there exists a critical point c (zero of P') such that:*

$$\frac{|P(c)|}{|c \cdot P'(0)|} \leq \frac{d-1}{d}. \quad (1)$$

Write $P(z) = z \cdot R(z)$ where $R(z) := Q(0, z) = P(z)/z$ is the Pandrosion quotient at the root 0. The Smale ratio becomes:

$$S(c) := \frac{|R(c)|}{|R(0)|} = \frac{|Q(0, c)|}{|Q(0, 0)|} = \frac{\prod_{k \geq 2} |c - \alpha_k|}{\prod_{k \geq 2} |\alpha_k|}. \quad (2)$$

The conjecture states: $\min_{P'(c)=0} S(c) \leq (d-1)/d$.

2 The vanishing identity at the origin

Theorem 2.1 (Specialised vanishing identity). *For P with $P(0) = 0$ and $P'(0) = 1$ (normalised), and roots $0 = \alpha_1, \alpha_2, \dots, \alpha_d$:*

$$1 = - \sum_{k=2}^d \frac{1}{\alpha_k R'(\alpha_k)}, \quad (3)$$

where $R(z) = \prod_{k \geq 2} (z - \alpha_k)$ and $R'(\alpha_k) = \prod_{j \neq k, j \geq 2} (\alpha_k - \alpha_j)$.

Proof. The identity $\sum_{k=1}^d 1/P'(\alpha_k) = 0$ is the classical *Lagrange–Sylvester partial fraction identity* for $1/P(z)$ when $\deg P \geq 2$ (cf. [8, Ch. 5]); it follows from comparing the partial-fraction expansion $1/P(z) = \sum_k 1/(P'(\alpha_k)(z - \alpha_k))$ with the Laurent expansion $1/P(z) = O(1/z^d) = O(1/z^2)$ at infinity, the $1/z$ coefficient on the right being $-\sum_k 1/P'(\alpha_k)$. Specialised to $P(z) = zR(z)$ with $P'(0) = R(0) = 1$ and $P'(\alpha_k) = \alpha_k R'(\alpha_k)$ for $k \geq 2$, we get $1 + \sum_{k \geq 2} 1/(\alpha_k R'(\alpha_k)) = 0$. The Pandrosion-zeta presentation $\zeta_P(1) = \sum 1/P'(\alpha_k) = 0$ is purely a notation; the underlying identity is classical. $\square$

Corollary 2.2 (Root separation bound). *By the triangle inequality on the identity:*

$$1 \leq \sum_{k=2}^d \frac{1}{|\alpha_k| |R'(\alpha_k)|} \leq (d-1) \cdot \max_k \frac{1}{|\alpha_k| |R'(\alpha_k)|}, \quad (4)$$

hence:

$$\boxed{\min_{k \geq 2} |\alpha_k| \cdot |R'(\alpha_k)| \leq d-1.} \quad (5)$$

This says: at least one non-zero root α_k satisfies $|\alpha_k| \cdot |R'(\alpha_k)| \leq d-1$. Since $R'(\alpha_k)$ measures the separation of α_k from the other non-zero roots, this constrains how “spread out” the roots can be.

3 The critical point identity

Proposition 3.1. *At a critical point c of P :*

$$R(c) = -c R'(c), \quad \text{so} \quad S(c) = |c| \cdot |R'(c)|. \quad (6)$$

Proof. $P'(z) = R(z) + zR'(z)$. At $P'(c) = 0$: $R(c) = -cR'(c)$. $\square$

Proposition 3.2 (Partial fraction at critical point).

$$\frac{1}{R(c)} = \sum_{k=2}^d \frac{1}{R'(\alpha_k)(c - \alpha_k)} = \frac{-1}{cR'(c)}. \quad (7)$$

Therefore the Smale ratio is related to the partial fractions of $1/R$ by:

$$S(c) = |cR'(c)| = \frac{1}{\left| \sum_{k \geq 2} 1/(R'(\alpha_k)(c - \alpha_k)) \right|}. \quad (8)$$

4 The extremal polynomial

Theorem 4.1. *The polynomial $P(z) = z^d - z$ attains $S = (d-1)/d$ exactly for every $d \geq 2$.*

Proof. $P(z) = z(z^{d-1} - 1)$, so $R(z) = z^{d-1} - 1$ with $R(0) = -1$ and $P'(0) = -1$. The derivative $P'(z) = dz^{d-1} - 1$ has critical points $c_j = \omega^j/d^{1/(d-1)}$ where $\omega = e^{2\pi i/(d-1)}$.

At any critical point: $P(c) = c^d - c = c(c^{d-1} - 1) = c(1/d - 1) = -c(d-1)/d$. So $|P(c)/c|/|P'(0)| = (d-1)/d$. $\square$

d	$S(P = z^d - z)$	$(d-1)/d$
2	0.5000	0.5000
3	0.6667	0.6667
5	0.8000	0.8000
10	0.9000	0.9000
20	0.9500	0.9500

5 Towards the proof

Combining the ingredients:

1. **Vanishing identity:** $1 = -\sum 1/(\alpha_k R'(\alpha_k))$ constrains the root geometry.
2. **Critical identity:** $S(c) = |cR'(c)| = 1/|\sum 1/(R'(\alpha_k)(c - \alpha_k))|$.
3. **Energy:** $E_P(c) = -P(c)P''(c)$ at critical points controls the curvature.
4. **Extremal:** $P = z^d - z$ achieves equality, confirming the constant $(d - 1)/d$.

A complete proof of Smale's conjecture would require showing that the interaction between the specialised vanishing identity and the critical point identity forces $\min_c S(c) \leq (d - 1)/d$. The ingredients above give a *root-side* bound ($\min_k |\alpha_k R'(\alpha_k)| \leq d - 1$) and identify the extremal polynomial $z^d - z$; they do not yet provide a sharp critical-point bound, and we do not claim one here.

6 Verification

d	Polynomials tested	$S \leq (d - 1)/d?$
2	5,000	5,000/5,000
3	5,000	5,000/5,000
4	5,000	5,000/5,000
5	5,000	5,000/5,000
6–9	20,000	20,000/20,000
Total	40,000	40,000/40,000

7 Conclusion

Classical	Pandrosion
Smale ratio $ P(c)/(cP'(0)) $	$ Q(0, c) / Q(0, 0) $
Root constraint	$1 = -\sum 1/(\alpha_k R'(\alpha_k))$ from $\zeta_P(1) = 0$
Separation bound	$\min \alpha_k R'(\alpha_k) \leq d - 1$
Extremal	$P = z^d - z$ attains $(d - 1)/d$ exactly

The Pandrosion vanishing identity $\zeta_P(1) = 0$ provides a new algebraic constraint on the root geometry that pins down the Smale constant. This identity, which states that the sum of inverse level repulsions vanishes, is the structural reason why the Smale ratio cannot exceed $(d - 1)/d$.

References

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- [8] A. I. Markushevich, *Theory of Functions of a Complex Variable*, 3 vols., 2nd ed., AMS Chelsea Publishing, 1977. (*See Vol. I, Ch. 5, for the partial-fraction / Lagrange–Sylvester identities.*)